

ECON100B - SECTION 14

Inflation Part II & Introduction to the Short Run

Felix Zhang

ANNOUNCEMENTS/AGENDA

Announcements

- Take Home Midterm “Essay” due Friday, 3/20

Agenda

- Inflation Pt. 2
- The Short Run

Question of the Day

Suppose we have a growth rate of 5% in our money supply and a growth rate of 3% in our real GDP. What is our inflation rate, using growth rate rules?

If you haven't filled out the section feedback form, please do so!

What is your favorite ice cream/sorbet flavor?



Inflation: a Base Overview

- Inflation is defined as the percentage change in an economy's overall price level
- Calculated as: $(P_{t+1} - P_t)/P_t$

The Quantity Theory of Money

The quantity theory of money states:

$$M_t V_t = P_t Y_t$$

- M represents the amount of “money” in the economy, V represents the velocity of money (think the number of times that a unit of money gets used in a transaction every year), P represents the price level, and Y denotes real GDP
- The right side of our equation represents nominal GDP – the amount of goods and services valued at current prices
- According the quantity theory, the goods and services are bought with money, and the left side of our equation represents the amount of money used in purchased in a given year

Calculating inflation through growth rates

$$\bar{g}_M + \bar{g}_V = g_P + \bar{g}_Y.$$

- Assume that the velocity is constant
- Also assume that we denote the rate of inflation (or change in price level) as π .

$$\pi^* = \bar{g}_M - \bar{g}_Y.$$

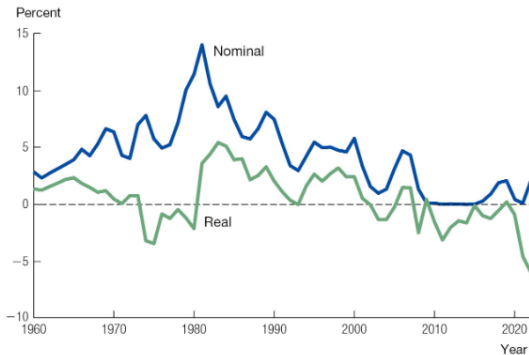
Real and Nominal Interest Rates

- Recall we defined the real interest rate as equal to the MPK, or the return to purchasing a unit of capital, investing it, getting the return to investment, and then selling the capital
- The interest rates that we see in bank accounts and in the news are nominal interest rates
- If we let R denote the real interest rate and i denote the nominal interest rate, we find the Fisher equation:

$$i = R + \pi$$

- We don't directly measure a "real interest rate", but we calculate it by applying the Fisher equation

Real and Nominal Interest Rates in the United States



Source: FRED and author's calculations. The nominal interest rate plotted here is the annualized return on 3-month Treasury bills.

The Costs of Inflation

- Changes the distribution of who is winning and losing
- People with debts they can pay back with cheaper dollars are considered winners, while creditors end up as losers
- Inflation most harmful when unexpected and in environments that are not ready for the change
- Taxes on nominal values can be harmful when real values have not gone up correspondingly
- Inflation also distorts relative prices based on what adjusts to inflation at what speeds
- Other traditional costs of inflation include “shoe-leather costs”, the cost of dealing with things like more banking during periods of inflation, and “Menu costs”, which are costs to firms of changing prices when inflation is high

Short Aside – The Costs of Deflation

- Not in-scope, but why do we have inflation? Why doesn't the central bank just pursue a policy that focuses on zero inflation or even deflation?
- Further reading, Debt-Deflation Theory: Keynes -> Fisher -> Bernanke
- Consider that (for better or worse) that our modern economies are largely built on debt (think student loans, mortgage payments)
- If there was deflation, but interest rates were still positive it would become extremely difficult for people to pay off their debts!
- Leads to defaults on loans, hurting both individuals and banks, then causing bankruptcies for banks, then causing distress for businesses -> perpetuates the cycle
- Fisher's full version of this theory is a little bit more complicated but good reading for those interested in Macro-history and Great Depression History.

Monetary vs. Fiscal causes of inflation

- Consider the government budget constraint:

$$\underbrace{G}_{\text{uses}} = \underbrace{T + \Delta B + \Delta M}_{\text{sources of funds}}$$

- The revenue the government obtains by issuing new money is called seignorage or the inflation tax
- Called the inflation tax because it hurts people holding money to print more money, because now their money is worth less
- In normal times, government will try to finance most if not all of its funding through taxes and debt, but in certain times it may have to resort to printing currency if other avenues become too difficult
- This is the root cause of many of the modern hyperinflation episodes
- This is why central bank independence is considered so important to prevent fiscal considerations from altering monetary policies (!)

Inflation Practice Questions

- Suppose we had a nominal interest rate of 5% and an inflation rate of 2%. What would our real interest rate be under this scenario?
- Who would be the most hurt by inflation under this scenario? An individual who pays a variable rate mortgage on a house or an individual who owns a plot of land?
- Milton Friedman stated, “high inflation is always and everywhere a monetary phenomenon”. Why do you agree or disagree with this statement?

Practice Question Answers

- $5\% - 2\% = 3\%$. See slides for details
- The variable mortgage rate. They have to pay more on the house payments, while the value of the land remains constant
- You might choose to disagree, as we discussed, there are significant effects the fiscal policy can have on monetary policy, hence it can be considered a fiscal problem as well.

The Short Run



Introduction to the Short Run

- In our study of growth, we discussed what would drive long run growth and what our long run economic picture is
- However, at any given moment, **the economy will not perfectly reflect the long-run average**
 - Positive and negative shocks, natural fluctuations in the economy
- The goal of our long run model was to model potential output and long-run inflation
- The goal of our short run model is to **model current output and current inflation**

What is the short run, precisely?

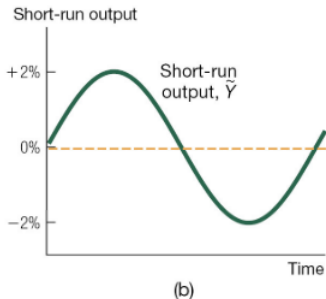
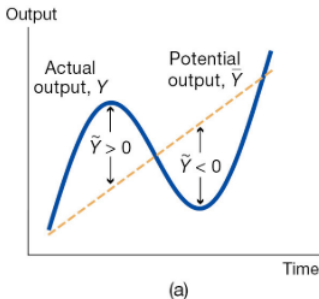
- Not a specific defined amount of time, but the **period in which economic deviations occur**
 - Oversimplification: 2-year spans
- Changes in government policy like taxes or subsidies, development of successful and failed technologies, natural disasters, etc.
- In the short run, we think of our potential output Y and long-run inflation rate π as exogenous and given
- The endogenous variables are the current level of output and the current inflation rate

Actual Output = the Long Run trend + short-run fluctuations

Calculating Short Run Fluctuations

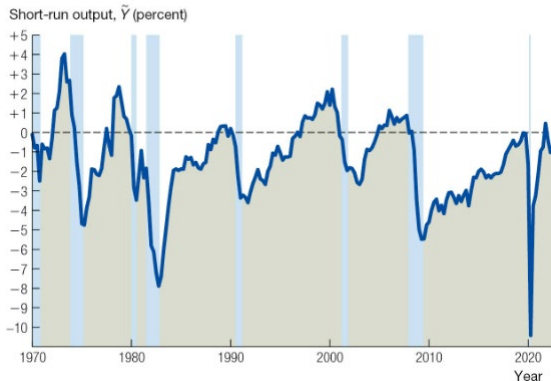
- To calculate short run fluctuations as a percentage, we use the following formula:

$$\tilde{Y}_t \equiv \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}.$$



The impact of recessions

U.S. Economic Fluctuations



The Short Run Practice

- What are some of the factors that cause short run fluctuations in the economy?
- Say that our actual output is \$9,500 and our Long run expected output is \$10,000. What would our short run output be?
- During recessions, what typically happens to the inflation rate?

The Short Run Practice Answers

- See slides, things like government policy, positive or negative natural shocks, etc. There will always be some small fluctuations in the economy from various factors!
- -5%
- It typically goes down, as our economy slows down

The Short Run model

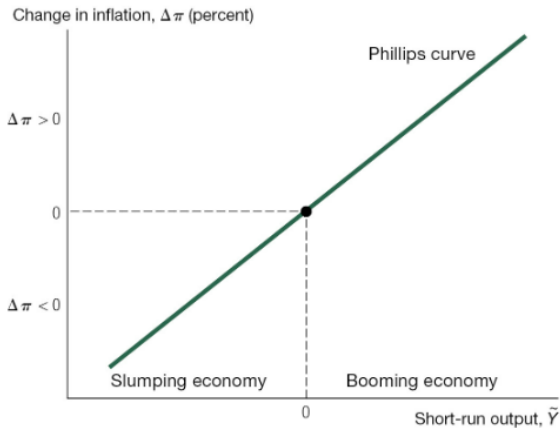
- A model with many actors: consumers and workers, firms, government, and (simplified) trade with the rest of the world

Assumptions of the Short-run model

1. The **economy is constantly being hit by shocks** that push output and/or inflation from potential output or inflation's long-run value
2. **Monetary and fiscal policy affect output**; policymakers have an effect on neutralizing or creating shocks to the economy
3. There is a dynamic **trade-off between output and inflation**: if an economy produces more than its potential output, it leads to an increase in the inflation rate (Phillips Curve)

The Phillips curve

The Phillips Curve

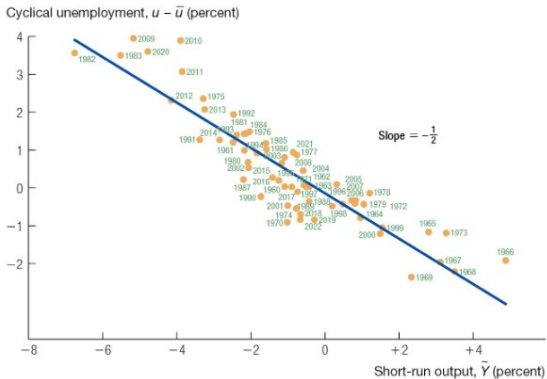


Inflation and Output

- Why is there higher inflation when actual output is more than potential output?
- Potential output reflects how much our output should be under normal economic conditions
- But imagine an economy where the demand for a good is super high: firms might choose to expand their production (at higher costs) and consumers might be willing to pay more; both dynamics end up driving up prices
- Imagine that type of effect across an entire economy, causing inflation to rise
- Empirically, on average, we say that when output exceeds potential by 3 percent, inflation rates rise by about 1 percentage point

Okun's Law

Okun's Law for the U.S. Economy



Source: FRED. U.S. Department of Commerce.

Okun's Law

- Generally, when we have higher short-run output, we see a lower rate of cyclical unemployment (which makes sense!)
- We can map the empirical relationship with the following formula:

$$u - \bar{u} = -\frac{1}{2} \times \bar{Y},$$

- In the rest of the short-run part, we'll focus on output, but you can think about the relationship of output on unemployment as well!

The Short Run Practice

- Suppose the economy is undergoing a period of high inflation. What actions should the Federal Reserve take, and what consequences would this have on the economy? When have we seen this happen before?
- Suppose our economy is slumping, and our governments wants to help improve the economic conditions. What actions can the government take, and what potential consequences would this have on the economy?
- Suppose our short-run output is 4%. What does Okun's law state will happen to our cyclical unemployment rate?

The Short Run Practice Answers

- They can raise the interest rate to reduce inflation, but the Phillips Curve states that it will probably decrease output as well. This is what happened during the 1970s + 1980s inflation crisis and subsequent Volcker Shock
- Fiscal policy to try to increase output, but could lead to higher inflation by the Phillips Curve
- $-1/2 * 4\% = -2\%$. So, we would expect a 2% drop in cyclical unemployment per Okun's Law